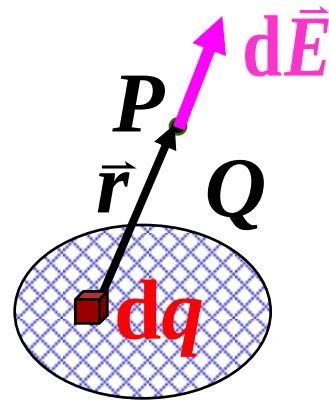


二、电荷连续分布的带电体的场强



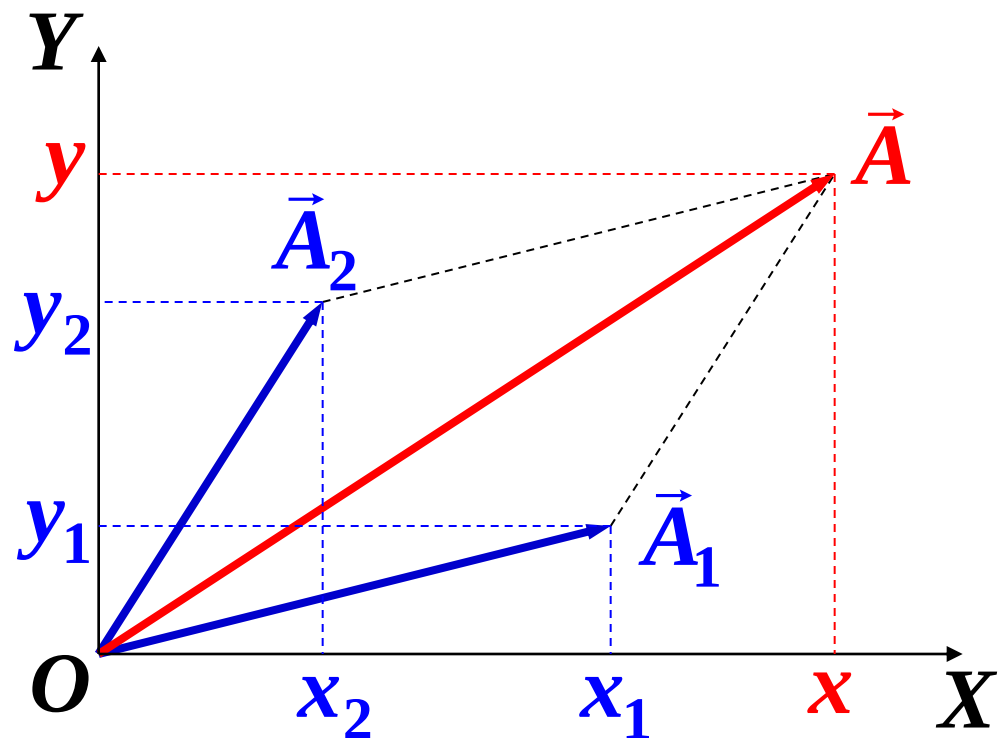
$$d\vec{E} = \frac{dq}{4\pi\epsilon_0 r^2} \hat{e}_r$$

$$\vec{E} = \int_{(Q)} d\vec{E} = \int_{(Q)} \frac{dq}{4\pi\epsilon_0 r^2} \hat{e}_r$$

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k} \rightarrow \begin{cases} E_x = \int_V dE_x \\ E_y = \int_V dE_y \\ E_z = \int_V dE_z \end{cases}$$

$$\vec{A}_1 = \hat{i} x_1 + \hat{j} y_1$$

$$\vec{A}_2 = \hat{i} x_2 + \hat{j} y_2$$



$$\vec{A} = \vec{A}_1 + \vec{A}_2 = \hat{i} (x_1 + x_2) + \hat{j} (y_1 + y_2) = \hat{i} x + \hat{j} y$$

$$\vec{A}_i = \hat{i} x_i + \hat{j} y_i + \hat{k} z_i$$

$$\vec{A} = \sum_i \vec{A}_i = \hat{i} \sum_i x_i + \hat{j} \sum_i y_i + \hat{k} \sum_i z_i = \hat{i} A_x + \hat{j} A_y + \hat{k} A_z$$

电荷密度

★ 体电荷密度

$$\rho = \frac{dq}{dV}$$

$$dq = \rho dV$$

★ 面电荷密度

$$\sigma = \frac{dq}{dS}$$



$$dq = \sigma dS$$

★ 线电荷密度

$$\lambda = \frac{dq}{dl}$$



$$dq = \lambda dl$$

[例] 均匀带电棒 (η , l) 的场强

$$dE_x = dE \sin \alpha = \frac{1}{4\pi\epsilon_0} \frac{\eta dy}{r^2} \sin \alpha$$

$$E_x = \int dE_x = \frac{\eta}{4\pi\epsilon_0} \int \frac{\sin \alpha}{r^2} dy$$

$$y = -x \operatorname{ctg} \alpha$$

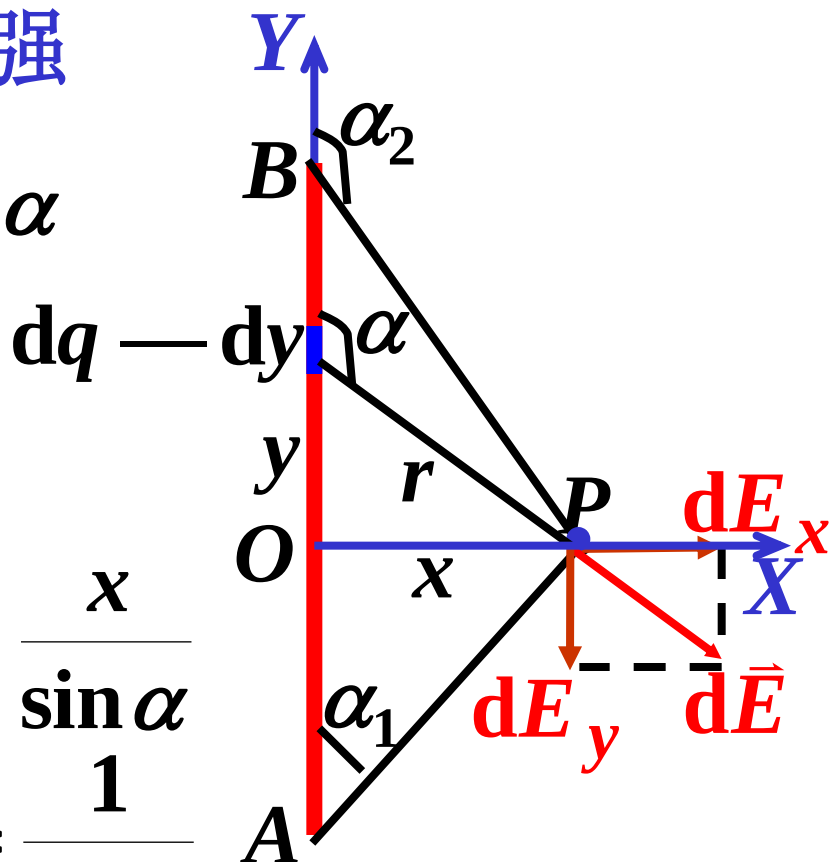
$$dy = x \csc^2 \alpha d\alpha$$

$$r = \frac{x}{\sin \alpha}$$

$$\csc \alpha = \frac{1}{\sin \alpha}$$

$$E_x = \frac{\eta}{4\pi\epsilon_0 x} \int_{\alpha_1}^{\alpha_2} \sin \alpha d\alpha = \frac{\eta}{4\pi\epsilon_0 x} (\cos \alpha_1 - \cos \alpha_2)$$

$$dE_y = dE \cos \alpha \quad E_y = \frac{\eta}{4\pi\epsilon_0 x} (\sin \alpha_2 - \sin \alpha_1)$$



[例] 均匀带电棒
(η , l) 的场强

$$E_x = \frac{\eta}{4\pi\epsilon_0 x} (\cos\alpha_1 - \cos\alpha_2)$$

$$E_y = \frac{\eta}{4\pi\epsilon_0 x} (\sin\alpha_2 - \sin\alpha_1)$$

讨论

1. P 点在中垂线上

$$\alpha_2 = \pi - \alpha_1$$

$$E_x = \frac{\eta \cos\alpha_1}{2\pi\epsilon_0 x} \quad E_y = 0$$

$x \gg l$ —— 点电荷场 $\cos\alpha_1 \approx \frac{l}{2x}$

$$E = \frac{\eta l}{4\pi\epsilon_0 x^2} = \frac{q}{4\pi\epsilon_0 x^2}$$

2. $l \rightarrow \infty$ —— 无限长 $\alpha_1 = 0 \quad \alpha_2 = \pi$

$$E_x = \frac{\eta}{2\pi\epsilon_0 x} \quad E_y = 0$$



$$\begin{aligned}
 r &= \frac{\eta}{2\pi\epsilon_0 E} \\
 &= \frac{1 \times 10^{-3} \text{ C/m}}{(2\pi)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(3 \times 10^6 \text{ N/C})} = 6 \text{ m}
 \end{aligned}$$

小结

积分法求场强的步骤

1. 选好微元，画出 $d\vec{E}$
2. 引入密度，写出 dE
3. 建立坐标，写出分量式
4. 统一变量，写出积分式
5. 定好上下限，注意对称性
6. 积分求结果，代数求数值
7. 进行讨论，加深理解。

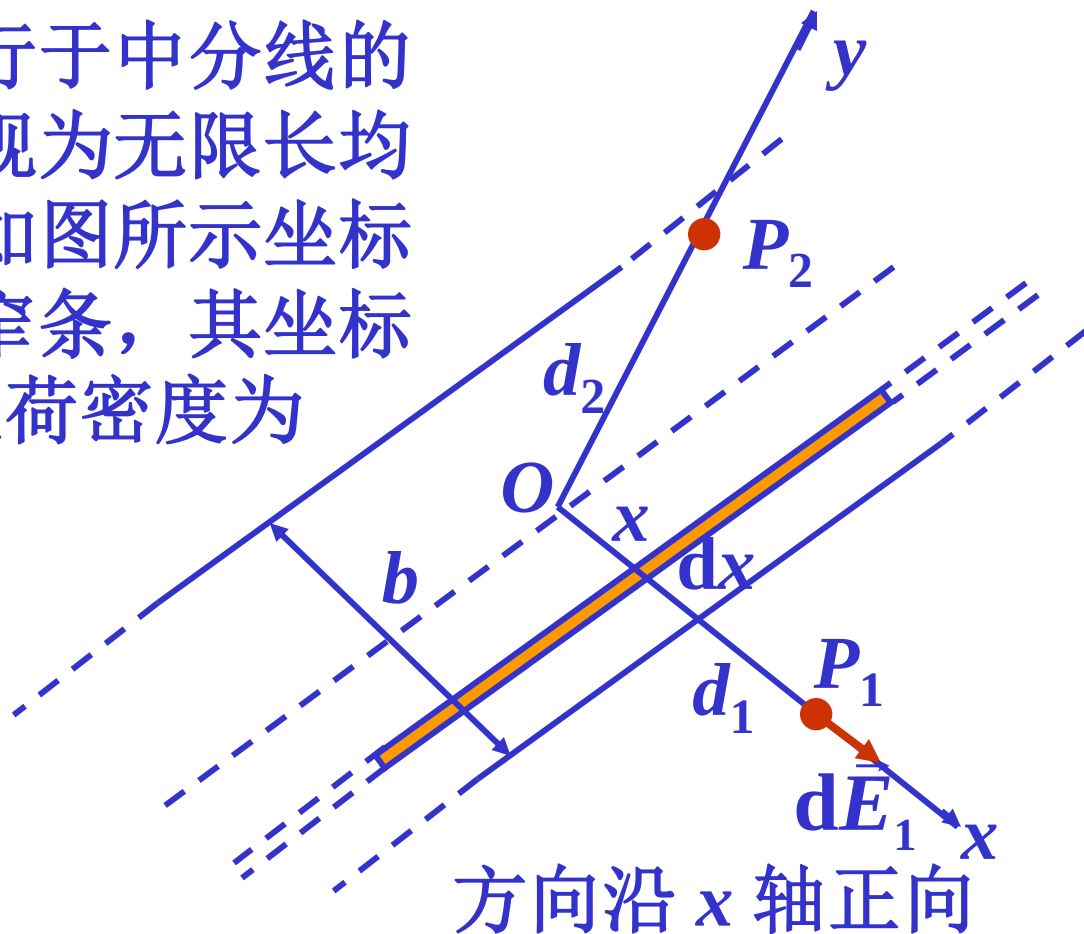
[例] 真空中有一无限长宽为 b 的薄平板，均匀带电，面电荷密度为 $\sigma (\sigma > 0)$ 。求：(1) 与平板共面且到平板中分线的距离为 d_1 ($d_1 > b/2$) 的 P_1 点的场强。(2) 过中分线的垂线上到平板距离为 d_2 的 P_2 点的场强。（习题7-6）

解：把平板分成许多平行于中分线的窄条，每个窄条都视为无限长均匀带电直线。建立如图所示坐标系，考虑其中一个窄条，其坐标为 x ，宽为 dx ，线电荷密度为

$$\lambda = \sigma dx$$

(1) 在 P_1 点，该电荷元产生的场强为

$$dE_1 = \frac{\sigma dx}{2\pi\epsilon_0(d_1 - x)}$$

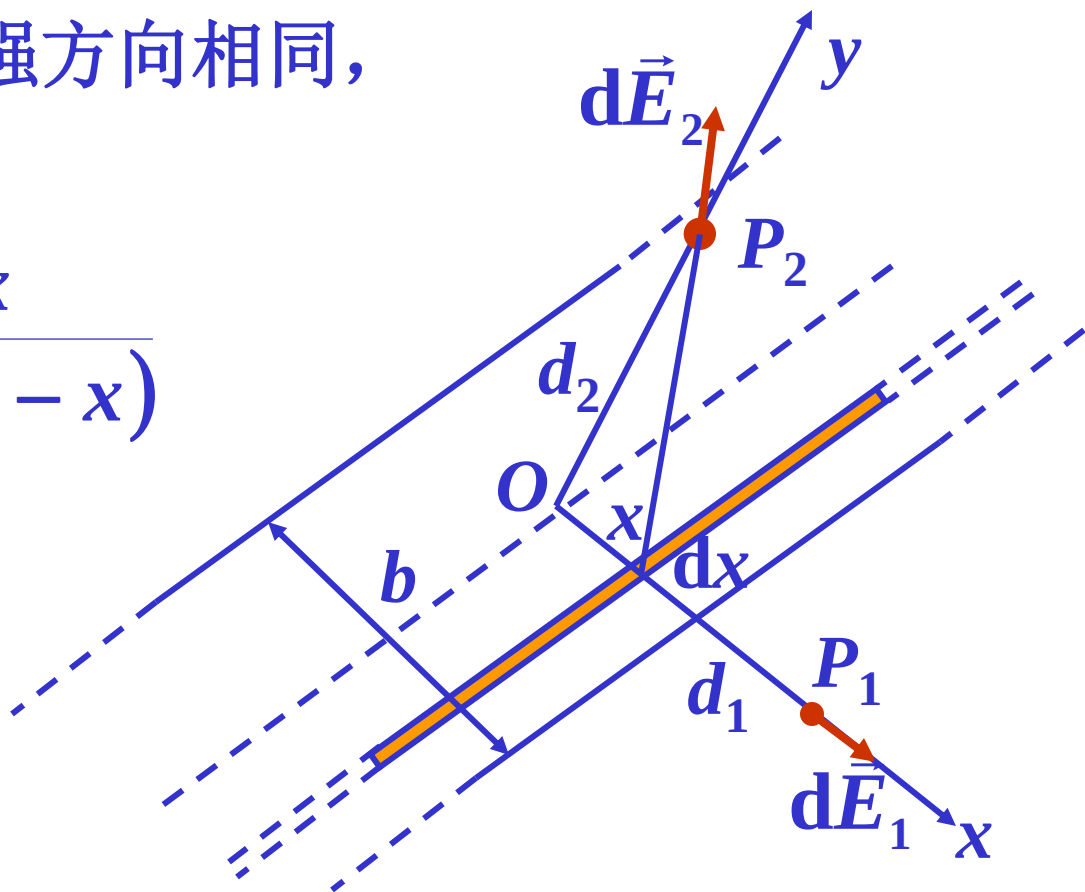


不同电荷元产生的场强方向相同，
所以不用分解

$$E_1 = \int dE_1 = \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{\sigma dx}{2\pi\epsilon_0(d_1 - x)}$$

$$= \frac{\sigma}{2\pi\epsilon_0} \ln \frac{2d_1 + b}{2d_1 - b}$$

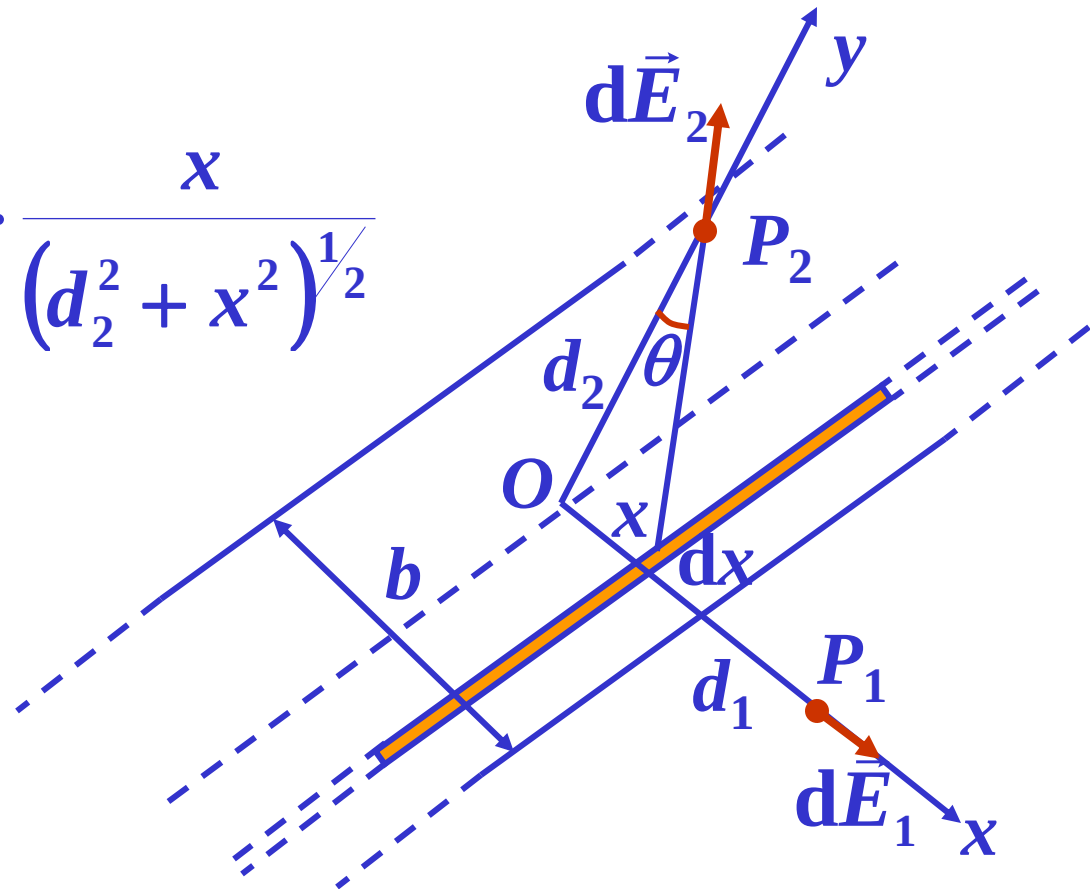
方向沿 x 轴正向



$$(2) \quad dE_2 = \frac{\sigma dx}{2\pi\epsilon_0(d_2^2 + x^2)^{1/2}} \quad \text{方向如图}$$

不同电荷元产生的场强方向不同，所以需要分解

$$\begin{aligned}
 dE_{2x} &= -dE_2 \sin \theta \\
 &= \frac{-\sigma dx}{2\pi\epsilon_0(d_2^2 + x^2)^{1/2}} \cdot \frac{x}{(d_2^2 + x^2)^{1/2}} \\
 &= \frac{-\sigma x dx}{2\pi\epsilon_0(d_2^2 + x^2)}
 \end{aligned}$$



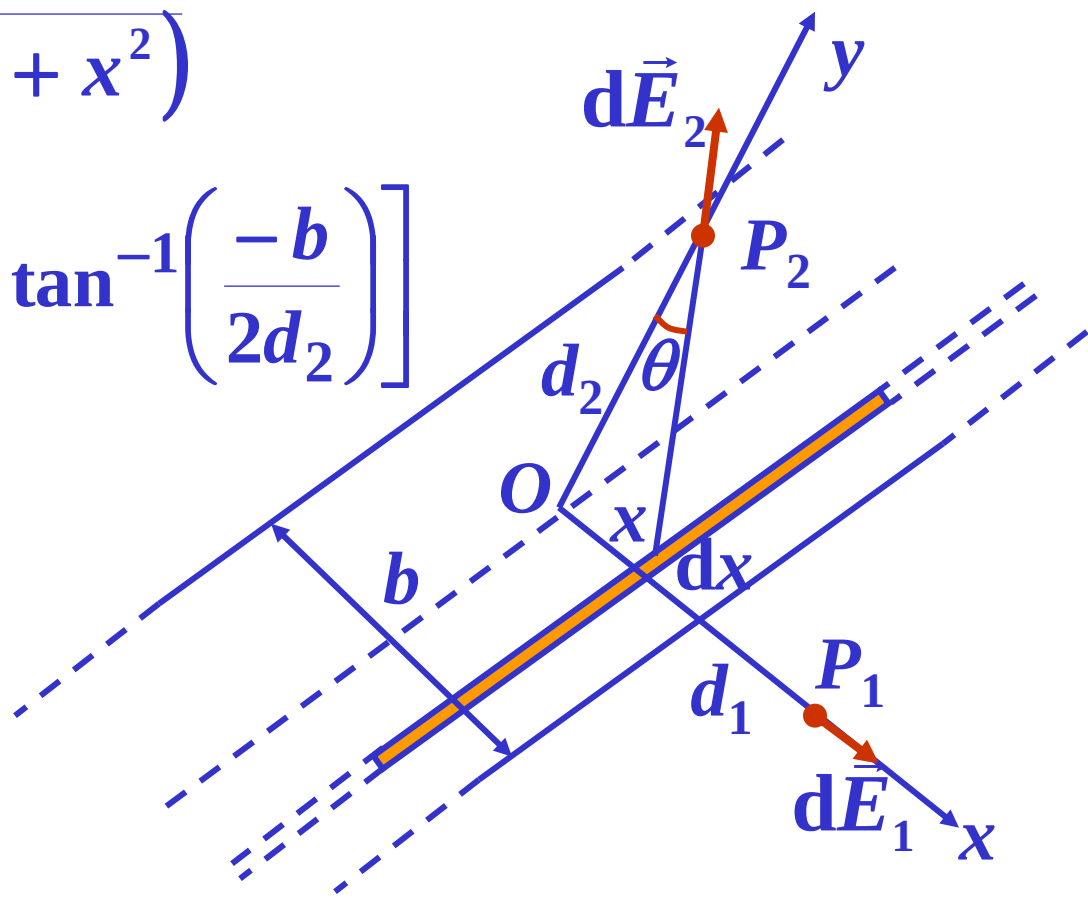
$$\begin{aligned}
 dE_{2y} &= dE_2 \cos \theta \\
 &= \frac{\sigma dx}{2\pi\epsilon_0(d_2^2 + x^2)^{1/2}} \cdot \frac{d_2}{(d_2^2 + x^2)^{1/2}} \\
 &= \frac{\sigma d_2 dx}{2\pi\epsilon_0(d_2^2 + x^2)}
 \end{aligned}$$

$$E_{2x} = \int dE_{2x} = \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{-\sigma x dx}{2\pi\epsilon_0(d_2^2 + x^2)} = 0$$

$$E_{2y} = \int dE_{2y} = \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{\sigma d_2 dx}{2\pi\epsilon_0(d_2^2 + x^2)}$$

$$= \frac{\sigma}{2\pi\epsilon_0} \left[\tan^{-1}\left(\frac{b}{2d_2}\right) - \tan^{-1}\left(\frac{-b}{2d_2}\right) \right]$$

也可由电荷元分布的对称性得到，请大家自己分析



讨论 (1) 对 P_2 点, 当 $b \gg d_2$ 时, 即无限大均匀带电平面

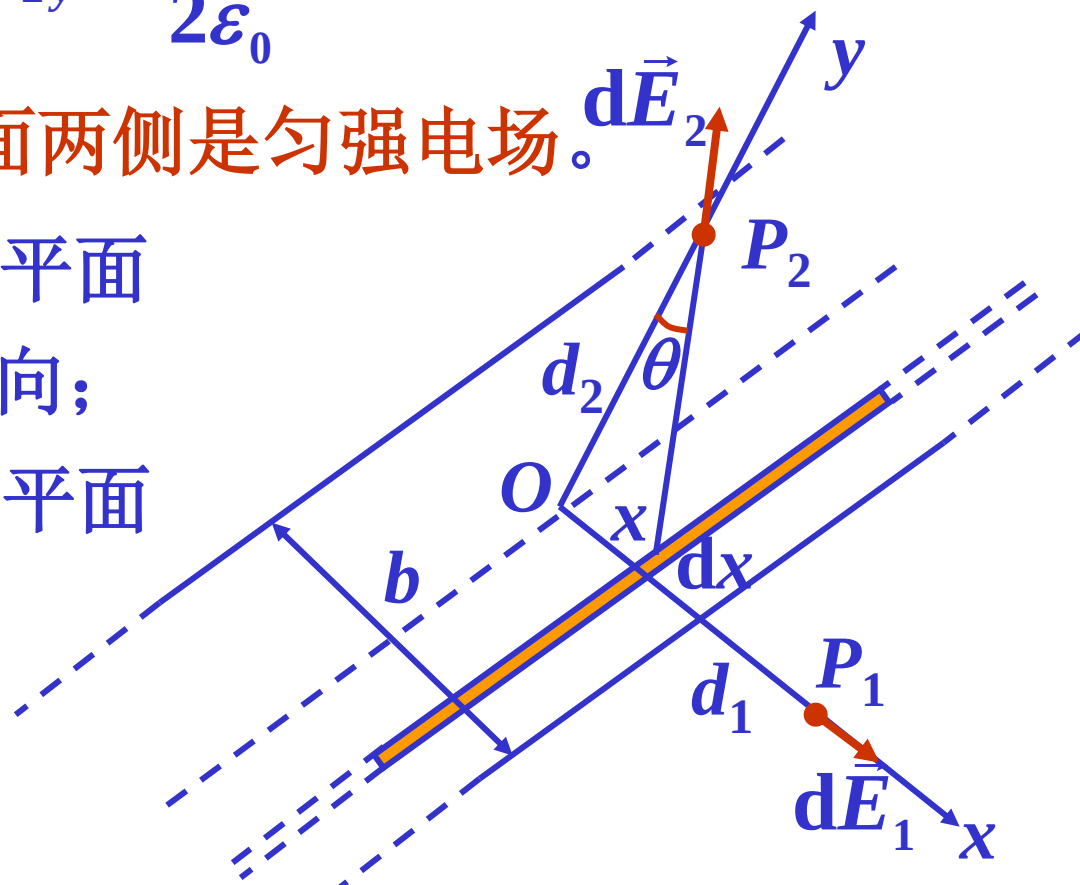
$$\tan^{-1}\left(\frac{b}{2d_2}\right) = \frac{\pi}{2}, \quad \tan^{-1}\left(\frac{-b}{2d_2}\right) = -\frac{\pi}{2}$$

$$E_2 = E_{2y} = \frac{\sigma}{2\varepsilon_0}$$

即: 无限大均匀带电平面两侧是匀强电场。

$\sigma > 0$, $\vec{E}$ 垂直带电平面
指向远离平面的方向;

$\sigma < 0$, $\vec{E}$ 垂直带电平面
指向平面;



讨论

(2) 对 P_1 点, 当 $d_1 \gg b$ 时

$$E_1 = \frac{\sigma}{2\pi\epsilon_0} \ln \frac{2d_1 + b}{2d_1 - b} = \frac{\sigma}{2\pi\epsilon_0} \ln \frac{1 + b/(2d_1)}{1 - b/(2d_1)}$$

$$|x| < 1$$

$$\ln(1+x)$$

$$\approx x - \frac{1}{2}x^2$$

$$= \frac{\sigma}{2\pi\epsilon_0} \left[\ln \left(1 + \frac{b}{2d_1} \right) - \ln \left(1 - \frac{b}{2d_1} \right) \right]$$

$$= \frac{\sigma}{2\pi\epsilon_0} \left[\frac{b}{2d_1} - \frac{1}{2} \left(\frac{b}{2d_1} \right)^2 + \frac{b}{2d_1} + \frac{1}{2} \left(\frac{b}{2d_1} \right)^2 \right]$$

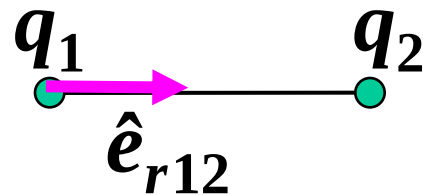
$$= \frac{\sigma b}{2\pi\epsilon_0 d_1} = \frac{\lambda}{2\pi\epsilon_0 d_1}$$

此时, 无限长均匀带电平板可视为无限长带电直线。

你记住了吗?

1. 库仑定律

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{e}_{r12}$$



2. 电力叠加原理

$$\vec{F} = \sum_i \vec{F}_i$$

3. 电场强度定义

$$\vec{E} = \frac{\vec{F}}{q_0}$$

单位: N/C or V/m

4. 点电荷的场强公式

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{e}_r$$

5. 场强叠加原理

$$\vec{E} = \sum_i \vec{E}_i$$

$$\vec{E} = \int d\vec{E}$$

记吧!